

MATHEMATICS

9709/42

Paper 4 Mechanics 1 (M1)

May/June 2014

1 hour 15 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



- 1 A car of mass 600 kg travels along a straight horizontal road. The resistance to the car's motion is constant and equal to R N.

(i) Find the value of R , given that the car's acceleration is 1.4 m s^{-2} at an instant when the car's speed is 18 m s^{-1} and its engine is working at a rate of 22.5 kW. [4]

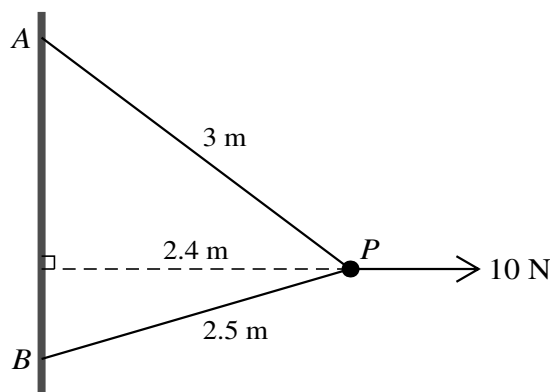
(ii) Find the rate of working of the car's engine when the car is moving with a constant speed of 15 m s^{-1} . [1]

- 2 A and B are two points which are 10 m apart on the same horizontal plane. A particle P starts to move from rest at A , directly towards B , with constant acceleration 0.5 m s^{-2} . Another particle Q is moving directly towards A with constant speed 0.75 m s^{-1} , and passes through B at the instant that P starts to move. At time T s after this instant, particles P and Q collide. Find

(i) the value of T , [4]

(ii) the speed of P immediately before the collision. [1]

3

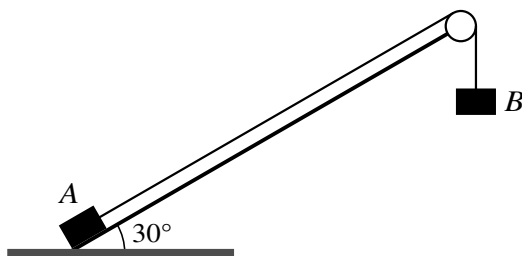


A and B are fixed points of a vertical wall with A vertically above B . A particle P of mass 0.7 kg is attached to A by a light inextensible string of length 3 m. P is also attached to B by a light inextensible string of length 2.5 m. P is maintained in equilibrium at a distance of 2.4 m from the wall by a horizontal force of magnitude 10 N acting on P (see diagram). Both strings are taut, and the 10 N force acts in the plane APB which is perpendicular to the wall. Find the tensions in the strings. [6]

- 4 A particle P moves on a straight line, starting from rest at a point O of the line. The time after P starts to move is t s, and the particle moves along the line with constant acceleration $\frac{1}{4} \text{ m s}^{-2}$ until it passes through a point A at time $t = 8$. After passing through A the velocity of P is $\frac{1}{2}t^{\frac{2}{3}} \text{ m s}^{-1}$.

(i) Find the acceleration of P immediately after it passes through A . Hence show that the acceleration of P decreases by $\frac{1}{12} \text{ m s}^{-2}$ as it passes through A . [4]

(ii) Find the distance moved by P from $t = 0$ to $t = 27$. [3]



A light inextensible rope has a block A of mass 5 kg attached at one end, and a block B of mass 16 kg attached at the other end. The rope passes over a smooth pulley which is fixed at the top of a rough plane inclined at an angle of 30° to the horizontal. Block A is held at rest at the bottom of the plane and block B hangs below the pulley (see diagram). The coefficient of friction between A and the plane is $\frac{1}{\sqrt{3}}$. Block A is released from rest and the system starts to move. When each of the blocks has moved a distance of $x\text{ m}$ each has speed $v\text{ m s}^{-1}$.

(i) Write down the gain in kinetic energy of the system in terms of v . [1]

(ii) Find, in terms of x ,

(a) the loss of gravitational potential energy of the system, [2]

(b) the work done against the frictional force. [3]

(iii) Show that $21v^2 = 220x$. [2]

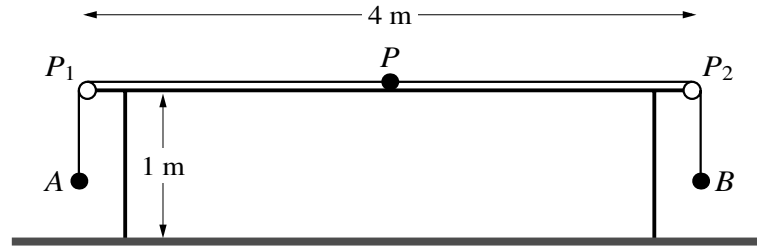
- 6 A particle P of mass 0.2 kg is released from rest at a point 7.2 m above the surface of the liquid in a container. P falls through the air and into the liquid. There is no air resistance and there is no instantaneous change of speed as P enters the liquid. When P is at a distance of 0.8 m below the surface of the liquid, P 's speed is 6 m s^{-1} . The only force on P due to the liquid is a constant resistance to motion of magnitude $R\text{ N}$.

(i) Find the deceleration of P while it is falling through the liquid, and hence find the value of R . [5]

The depth of the liquid in the container is 3.6 m . P is taken from the container and attached to one end of a light inextensible string. P is placed at the bottom of the container and then pulled vertically upwards with constant acceleration. The resistance to motion of $R\text{ N}$ continues to act. The particle reaches the surface 4 s after leaving the bottom of the container.

(ii) Find the tension in the string. [4]

[Question 7 is printed on the next page.]



A light inextensible string of length 5.28 m has particles A and B , of masses 0.25 kg and 0.75 kg respectively, attached to its ends. Another particle P , of mass 0.5 kg, is attached to the mid-point of the string. Two small smooth pulleys P_1 and P_2 are fixed at opposite ends of a rough horizontal table of length 4 m and height 1 m. The string passes over P_1 and P_2 with particle A held at rest vertically below P_1 , the string taut and B hanging freely below P_2 . Particle P is in contact with the table halfway between P_1 and P_2 (see diagram). The coefficient of friction between P and the table is 0.4. Particle A is released and the system starts to move with constant acceleration of magnitude $a \text{ m s}^{-2}$. The tension in the part AP of the string is $T_A \text{ N}$ and the tension in the part PB of the string is $T_B \text{ N}$.

- (i) Find T_A and T_B in terms of a . [3]
- (ii) Show by considering the motion of P that $a = 2$. [3]
- (iii) Find the speed of the particles immediately before B reaches the floor. [2]
- (iv) Find the deceleration of P immediately after B reaches the floor. [2]